

Pattern-generic Gershgorin reduction of the beyond-layer bound (gate STEP-5B)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-05 · this version issued 2026-06-05 · v1.3

1. Purpose and scope

Gate STEP-5B requires a *class-wide* bound: for every admissible condensate pattern (not merely the five enumerated readings), the beyond-layer correction — the off-diagonal Bloch channel discarded by the isotropic Gaussian–Hartree layer of hypothesis H-LAYER — must not overcome the layer margins of Proposition A (weakest interval margin +0.00432, the band figure of Math437 v1.2 after the Math440 precision fix). This note (i) proves two pattern-independent lemmas (Gershgorin–Schur row bound; second-order log-det envelope with an isotropic trace identity), (ii) states the resulting *precise reduction* of STEP-5B to two named gaps, and (iii) certifies all entering constants numerically at the production anchor, reproducing the Math434-audit calibration. It claims NEITHER closure of STEP-5B NOR any tier action on the Reading-H selection. The result is registered at T4 because Lemmas A/B/C'/D and the closed-region theorem are now derived (the T3 → T4 action of this re-issue is recorded in the footer); the class-wide statement remains open.

2. Setting

Work at the corrected-convention production anchor ($\mu^2 = 0.005$, $r_{\text{braz}} = K(q_0) = \mu^2$; claim A1-KERNEL-CONV), with the dressed isotropic diagonal

$$D(k) = \hat{r} + C(k^2 - q_0^2)^2, \quad \hat{r} = r_R + 2\lambda' I, \quad \lambda' = 3u + 30vM,$$

where $r_R = 0.3045257$, $M = M(\hat{r})$ self-consistently ($M_R = 0.109414$ at $I = 0$), $u = -0.86$, $v = +3.24$, $C = 1$, $q_0 = 0.6801748$, and $I = \sum_i A_i^2$ is the total condensate intensity of an admissible pattern $\phi_c(x) = \sum_i 2A_i \cos(q_i \cdot x + \varphi_i)$, $|q_i| = q_0$. Note $\lambda' = 8.0551 > 0$ at the anchor: the pattern Hartree term *raises* the diagonal. The beyond-layer channel is the off-diagonal part W of the Bloch Hessian in the plane-wave basis: entries w_t at momentum transfers $t \in T(Q) = \{\pm q_i \pm q_j\} \setminus \{0\}$ with quartic-vertex weights

$$w_t = \lambda' \sum_{(i,j): \pm q_i \pm q_j = t} A_i A_j$$

(sextic $5vA^4$ -class transfers enter at $O(w_4)$; gap G2). Define $X = D^{-1/2} W D^{-1/2}$ and the weighted transfer norm $W_{\ell^1}(P) = \sum_{t \neq 0} |w_t|$.

3. Two pattern-independent lemmas

Lemma 1 (Gershgorin–Schur row bound). *For the symmetric kernel X above, with $D_{\min} = \hat{r}$,*

$$\|X\| \leq \sup_k \sum_{t \neq 0} \frac{|w_t|}{\sqrt{D(k)D(k+t)}} \leq W_{\ell^1}(P) g(\hat{r}), \quad g(\hat{r}) = \frac{1}{\hat{r}} + \frac{1}{\sqrt{\hat{r}(\hat{r} + 64Cq_0^4)}}.$$

Proof. The first inequality is the Schur test with unit weights (equivalently the Gershgorin row bound for the symmetric matrix truncation of the Bloch problem; entries decay quartically in the band index, so the truncation limit is harmless). For the second: each transfer term is maximised when both legs are on-shell, giving $1/\hat{r}$; retaining additionally the degenerate-endpoint term $1/\sqrt{\hat{r}(\hat{r} + 64Cq_0^4)}$ of the Math434 LAM geometry makes g an upper envelope for every $|t| \leq 2q_0$ uniformly in the row, since any further leg sits at distance $\geq 64Cq_0^4$ above the gap in the quartic well. Summing over transfers yields the claim. $\square$

Lemma 2 (Second-order log-det envelope; isotropic trace identity). *If $\|X\| \leq a < 1$, the beyond-layer free-energy correction per unit volume obeys*

$$|\delta F_{\text{off}}| = \frac{1}{2V} |\text{tr}(\ln(\mathbf{1} + X) - X)| \leq \frac{\text{tr } X^2/V}{4(1-a)}, \quad \frac{\text{tr } X^2}{V} = \sum_{t \neq 0} |w_t|^2 J(|t|),$$

$$J(|t|) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{D(k)D(k+t)},$$

which depends on the pattern only through the moduli $|t|$ (isotropy of D).

Proof. Eigenvalues x_i of X lie in $[-a, a]$; the scalar Taylor remainder gives $|\ln(1+x) - x| \leq x^2/(2(1-a))$ on that interval; summing over the spectrum and dividing by $2V$ gives the first display (the $\frac{1}{2}$ from the Gaussian free energy). The trace identity is direct: $\text{tr } X^2 = \sum_k \sum_t w_t^2/(D(k)D(k+t)) \rightarrow V \sum_t w_t^2 J(|t|)$, and J depends only on $|t|$ because D is a function of $|k|$ alone. The first off-diagonal order $\text{tr } X$ vanishes (no $t = 0$ entry in W). $\square$

Lemma 3 (Transfer matching — exact ℓ^1 identity and caps). *For any admissible single-shell pattern ($Q = \{q_1, \dots, q_n\}$ distinct, amplitudes $A_i > 0$, $I = \sum_i A_i^2$, $S = \sum_i A_i$):*

- (i) *for every fixed transfer $t \neq 0$ and fixed sign pair, the pairs (i, j) with $\pm q_i \pm q_j = t$ form a partial matching (q_j is uniquely determined by q_i and t); hence the multiplicity obeys $m_t \leq 2n$ and, by Cauchy–Schwarz along the matching, $|w_t| \leq 2\lambda' I$;*
- (ii) *$\sum_{t \neq 0} |w_t| \leq \lambda' (4S^2 - 2I) \leq \lambda' (4nI - 2I)$, with equality in the first inequality iff no two ordered sign-pairs produce coinciding transfers (verified to machine precision on planar rings $n = 8..32$).*

Proof. (i) Uniqueness of $q_j = \pm(t \mp q_i)$ on the shell gives the matching; the weight along a matching is $\sum A_i A_{\sigma(i)} \leq \sqrt{\sum A_i^2} \sqrt{\sum A_{\sigma(i)}^2} \leq I$ per sign pair, and two sign pairs can land on the same t . (ii) Summing $A_i A_j$ over all ordered pairs and the four sign combinations counts $4S^2$, minus the excluded $t = 0$ diagonal contributions $2I$; Cauchy–Schwarz gives $S^2 \leq nI$. $\square$

Lemma 4 (ℓ^2 mass). $\sum_{t \neq 0} |w_t|^2 \leq (\max_t |w_t|) \sum_{t \neq 0} |w_t| \leq 2\lambda' I \cdot \lambda' (4S^2 - 2I) \leq 8n (\lambda' I)^2$.

Theorem 1 (Closed region of STEP-5B — derived). *Let $a(n, I) = \lambda'(4nI - 2I) g(\hat{r}(I))$. For every admissible single-shell pattern with mode count n and intensity I such that $a(n, I) \leq a_0 = 0.75$ and*

$$\delta(n, I) = \frac{2\lambda'I \cdot \lambda'(4nI - 2I) J_{\max}}{2(1 - a(n, I))} < 0.00432,$$

the beyond-layer correction cannot overcome the weakest layer margin. At the production anchor the region contains all patterns with

$$\begin{aligned} n \leq n_{\max}(I) : \quad & n_{\max}(10^{-4}) = 62, \quad n_{\max}(2 \times 10^{-4}) = 31, \quad n_{\max}(4 \times 10^{-4}) = 16, \\ & n_{\max}(10^{-3}) = 6, \quad n_{\max}(2 \times 10^{-3}) = 3. \end{aligned}$$

In particular every configuration in the v1.0 calibrated boxes is now DERIVED ($n = 12 \leq 16$ at $I = 4 \times 10^{-4}$, margin ratio ≥ 13).

Proof. Combine Lemmas A–D: $\|X\| \leq a(n, I)$ by Lemma A with the Lemma-C' ℓ^1 cap; Lemma B with the Lemma-D ℓ^2 mass gives $|\delta F_{\text{off}}| \leq \delta(n, I)$; the displayed inequality is then checked pointwise on the integer grid up to the stated $n_{\max}$ (script v1.1.1, 54/54 asserts; monotonicity of $n_{\max}$ in I asserted). $\square$

Lemma 5 (Sphere additive energy — diagonal/off-diagonal split). *Let $\hat{Q} = \{\pm q_i\}$ with amplitudes $a_{\pm q_i} = A_i$, participation factor $\varphi(P) = n \sum_i A_i^4 / I^2 \in [1, n]$ ($\varphi = 1$ for equal spread), and discrete nonzero-translate overlap $\nu^*(P) = \max_{c \neq 0} |\hat{Q} \cap (\hat{Q} + c)|$ over realized displacements $c = x - y'$, $x, y' \in \hat{Q}$. Then*

$$\sum_{t \neq 0} |w_t|^2 \leq 4\lambda'^2 I^2 (\varphi(P) + \nu^*(P)).$$

Proof. Write $w_t / \lambda' = (f * f)(t)$ with $f = \sum_{u \in \hat{Q}} a_u \delta_u$. Then $\sum_{t \neq 0} |w_t / \lambda'|^2$ is the count-weighted sum over ordered quadruples $x + y = x' + y' = t \neq 0$ in $\hat{Q}$ of $a_x a_y a_{x'} a_{y'}$. AM–GM symmetrisation bounds this by $\sum_{x, y'} a_x^2 a_{y'}^2 N(x, y')$, where $N(x, y') = \#\{x' \in \hat{Q} : x' - (x - y') \in \hat{Q}\} = |\hat{Q} \cap (\hat{Q} + (x - y'))|$. The diagonal pairs $x = y'$ give $N = |\hat{Q}| = 2n$ against $\sum_x a_x^4 = 2 \sum_i A_i^4$, i.e. $4\lambda'^2 I^2 \varphi$. The off-diagonal pairs have $N \leq \nu^*$ against $\sum_{x \neq y'} a_x^2 a_{y'}^2 \leq (2I)^2$. $\square$

Corollary 1 (Transversal n -free bound). *For every admissible single-shell pattern with $\nu^* \leq 4$ and $\varphi \leq 1 + \epsilon$ (in particular generic/transversal configurations: measured $\nu^* = 2$ on random shells $n = 12, 24$), the beyond-layer envelope is n -FREE: at the anchor, with $c_E = 4(\varphi + \nu^*) \leq 20$,*

$$|\delta F_{\text{off}}| \leq \frac{c_E (\lambda'I)^2 J_{\max}}{2(1 - a_E)} < 0.00432,$$

with margin ratios $131 \times 16 \times 2 \times$ at $I = 4 \times 10^{-4} / 10^{-3} / 2 \times 10^{-3}$, where the spectral control a_E uses the E -route operating envelope (at most $\nu^ + 2$ heavy transfers per row, each of weight $\leq 2\lambda'I$) — the row-count statement is the remaining $G1''$ item for this class.*

4. The reduction (precise form of STEP-5B)

Proposition 1 (Pattern-generic reduction). *Let P be any admissible pattern with $a(P) := W_{\ell^1}(P) g(\hat{r}(I)) < 1$. Then*

$$|\delta F_{\text{off}}(P)| \leq \frac{W_{\ell^1}(P)^2 J_{\max}}{4(1 - a(P))}, \quad J_{\max} := \max_{0 < |t| \leq 2q_0} J(|t|).$$

Consequently STEP-5B holds on the region $\{P : \text{RHS} < \text{interval margin}\}$, and full closure of the gate reduces to exactly two gaps:

- ***G1' residual — restructured by Lemma E:*** (a) transversal class ($\nu^* \leq 4, \varphi \lesssim 1$): the n -free ℓ^2 bound is DERIVED (Lemma E + corollary); what remains for full rigour here is only the row-count statement behind a_E (at most $\nu^* + 2$ heavy transfers per row) — item $G1''(\text{row})$. (b) ring/degenerate family ($\nu^* \sim n$ via the vertical antipodal displacement): the sup- ν^* route is intrinsically loose there, but the family is one-parameter and explicitly computable — measured $c(n) = 11.75 \rightarrow 13.72$ saturating over $n = 8.64, \leq 16$; the designated route is the rotation-orbit decomposition of the transfer set (each orbit contributes $O(1)$), to be written as the ring-family proposition. (c) decomposition glue ($G1''$): split a general pattern into a ν -transversal part plus circle-clustered parts and control the bilinear cross terms of W ; together with (a) and (b) this closes $G1'$.
- ***G2 (vertex bookkeeping completeness):*** sextic $O(w_4)$ -class transfers, two-shell cross transfers, and the $\sigma(x)$ -inhomogeneity channel (Math429 geometry) folded into the same envelope.

5. Numerical verification

Script [codes/vacuum/beyond_layer_gershgorin_bound.py](#) (v1.2.1; 69/69 self-test asserts PASS; artefact [runs/B5-BEYOND-LAYER-BOUND/260605-gershgorin-reduction/result.json](#)).

Quantity	Value	Check
anchor $r_R, M_R, \lambda', 64Cq_0^4$ Math434 calibration ($A = 0.01$): $\hat{r}'$, row terms	0.3045257, 0.109414, 8.0551, 13.698 0.30613; $2.624 \times 10^{-3} + 3.879 \times 10^{-4}$	match locked constants reproduces $2.633 \times 10^{-3} + 3.893 \times 10^{-4}$; $\ X\ \leq 3.1 \times 10^{-3}$
$J(t), t /q_0 \in \{0.5, 1, \sqrt{2}, \sqrt{3}, 2\}$	0.273, 0.218, 0.169, 0.133, 0.104	grid-refinement drift $< 6 \times 10^{-6}$; shell estimate 0.161 brackets $J_{\max}$
calibrated box $n_{\text{res}}=12, I = 4 \times 10^{-4}$	$a = 0.143, \delta \leq 2.38 \times 10^{-4}$	margin ratio 18.2×
calibrated box $n_{\text{res}}=12, I = 10^{-3}$	$a = 0.347, \delta \leq 1.95 \times 10^{-3}$	margin ratio 2.2×
LAM second order	$\delta_{\text{LAM}} = 6.7 \times 10^{-8}$	margin/64,000
matching lemmas on rings/random shells	exact ℓ^1 identity; $m_t \leq 2n$; $w_t \leq 2\lambda' I$	equality on rings to 10^{-12}
n -uniform ℓ^2 evidence (rings 16–32)	$\sum w_i^2 = 12.9\text{--}13.5 (\lambda' I)^2$	vs $8n$ bound: 19× slack at $n=32$
derived $n_{\max}(I)$	$62/31/16/6/3$ at $I = 10^{-4}..2 \times 10^{-3}$	$a \leq 0.75$; boxes of v1.0 now derived
Lemma E split (rings/random, $n \leq 64$)	$c_{\text{meas}} \leq 4(\varphi + \nu^*)$ everywhere	random: $\nu^* = 2$, $c_{\text{meas}} = 7.5\text{--}7.75 \leq 12$
ring family $c(n)$	$11.75 \rightarrow 13.72$ ($n = 8.64$), ≤ 16	saturating; orbit/family route
transversal n -free corollary	ratios $131 \times /16 \times /2 \times$ at $I = 4 \times 10^{-4} / 10^{-3} / 2 \times 10^{-3}$	$\nu^* \leq 4, \varphi \leq 1$; n -FREE

Quantitative sanity check (mandatory): the LAM row terms agree with the Math434-audit figures to $\leq 4 \times 10^{-6}$; J obeys the analytic shell estimate $q_0/(8\pi\hat{r}^{3/2}\sqrt{C}) = 0.161$ within a factor 1.7; dimensions: $[w] = [\hat{r}]$, $[J] = [\hat{r}]^{-2} [k]^3$ consistent with δF per unit volume.

The original v1.0 calibrated boxes are now DERIVED within the closed-region theorem; the remaining non-derived part is the thin-spread regime $n > n_{\max}(I)$, recorded as $G1'$.

6. Devil's-advocate (three objections)

- (α) “ W_{ℓ^1} grows without bound in n , so nothing class-wide is proven.” VALID-with-mitigation, now sharpened: the closed-region theorem converts the ℓ^1 route into an explicit derived region

$n \leq n_{\max}(I)$; the residual is exactly $G1'$ (thin-spread), for which the n -uniform ℓ^2 ring evidence identifies the designated attack. The class-wide statement remains open, hence T4 and not T5.

- (α') “*The author’s own constants may be wrong.*” Two exhibits, both caught by the verification loop before registration: (1) the v1.1.0 ℓ^1 assert with the wrong constant $2S^2$ FAILED on every configuration; ring measurements matched the corrected identity $\lambda'(4S^2 - 2I)$ to 10^{-12} . (2) the v1.2.0 circle count for Lemma E included the $c = 0$ diagonal ($\nu = 2n$ on every configuration); the failed transversal asserts exposed it, and the corrected diagonal/off-diagonal split passes 69/69.
- (β) “ *$J(|t|)$ is numerical only.*” DISMISSED with evidence: grid-refinement drift $< 6 \times 10^{-6}$ relative, UV tail quartically convergent, and the independent analytic shell estimate brackets $J_{\max}$ within a factor 1.7.
- (γ) “*The comparison uses the global minimum margin, but Proposition A’s floors are interval-wise.*” VALID-with-mitigation: using the weakest interval margin $+0.00432$ globally is conservative by construction; an interval-wise comparison can only enlarge the certified region.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND (serving gate STEP-5B)
Precise statement:	Lemmas A/B/C'/D (rigorous) + the derived closed-region theorem: STEP-5B holds for all single-shell patterns with $n \leq n_{\max}(I)$ (62/31/16/6/3 at $I = 1e-4..2e-3$); residual = $G1'$ (thin-spread) + $G2$ (vertex bookkeeping); 54/54 asserts at the production anchor.
Scope:	Reduction + calibrated-region certification only; STEP-5B remains OPEN; no tier action on B1/B2.
Dependencies:	A1-KERNEL-CONV (convention); margins from Proposition A (B2-PROPA-HLAYER chain).
Evidence grade:	ANALYTIC (lemmas) + EXECUTED (constants)
Reproduction command:	python codes/vacuum/beyond_layer_gershgorin_bound.py
Expected output:	claims 54/54 PASS; result.json under runs/B5-BEYOND-LAYER-BOUND/260605-gershgorin-reduction/
Falsification gate:	An admissible pattern inside a certified box whose exact off-diagonal correction exceeds the Lemma-B envelope, or any pattern undercutting the $+0.00432$ band margin at the anchor.
Tier before / after:	T3 / T4
No-overclaim statement:	This note does NOT close STEP-5B and does NOT promote Reading-H; class-wide validity awaits $G1' + G2$; the n -free l2 constant is EVIDENCE only.
Next required action:	$G1'$ -- prove the n -free l2 theorem ($\sum_t w_t^2 \leq c (\lambda I)^2$, ring evidence $c \sim 14$) + a second-moment spectral bound replacing l1.